College of Dentistry
3rd Stage
Prosthodontics II
2025-2026
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Lecture 11

Intracoronal Direct Retainers in RPD

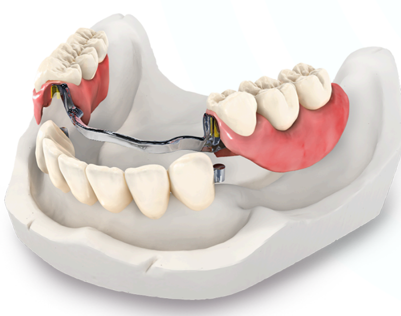
Learning Objectives

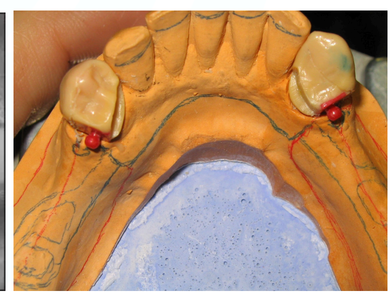
At the end of this lecture, students will be able to:

- Understand intracoronal direct retainers: their definition, components, advantages, disadvantages, and limitations.
- Differentiate between different classifications of intracoronal direct retainers.

Intracoronal Direct Retainers

It refers to an interlocking device, one component is fixed to an abutment or abutments, and the other is integrated into a removable prosthesis to stabilise and retain it. They are designed to replace the occlusal rest, bracing arm, and retaining arm of a conventional clasp unit



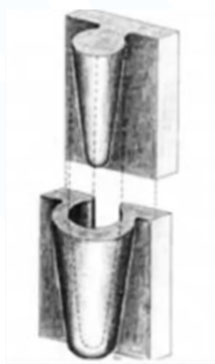


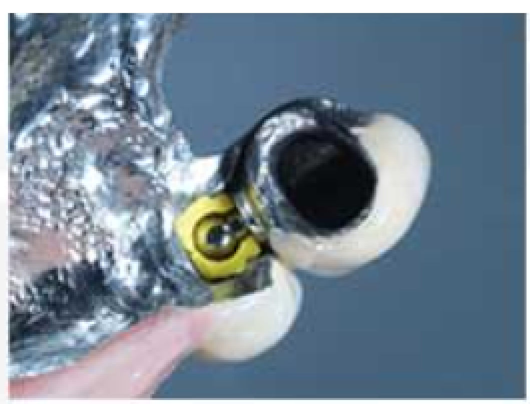
Classifications of intracoronal attachments:

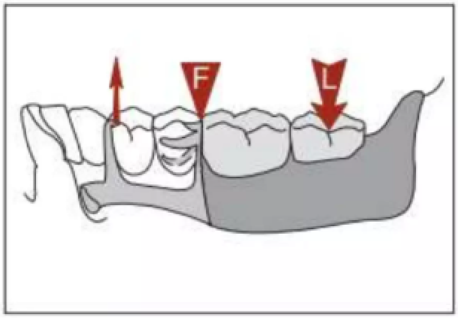
1. **According to the method of fabrication:**
   a. **Precision attachment:** a prefabricated device. It comprises a key and keyway, with opposing vertical parallel walls that limit movement and resist the removal of the partial denture through frictional resistance.
   b. **Semi-precision attachment:** a custom-made device. This category shows a less intimate fit between its components. Components usually originate as wax or plastic patterns, which are subsequently cast in metal.

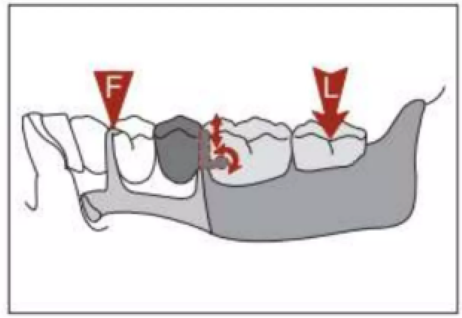
2. **According to their relation to the abutment tooth**
    a. **Internal attachments**: they are incorporated entirely within the contours of a cast crown.
    b. **External attachments**: they are situated external to the crown.


3. **According to the stiffness of the resulting joint**
    a. **Rigid attachment:** It is used as a retainer for tooth-borne situations, where vertical movement only is allowed.
    b. **Resilient attachment:** It is used as a stress breaker retainer in free-end situations.



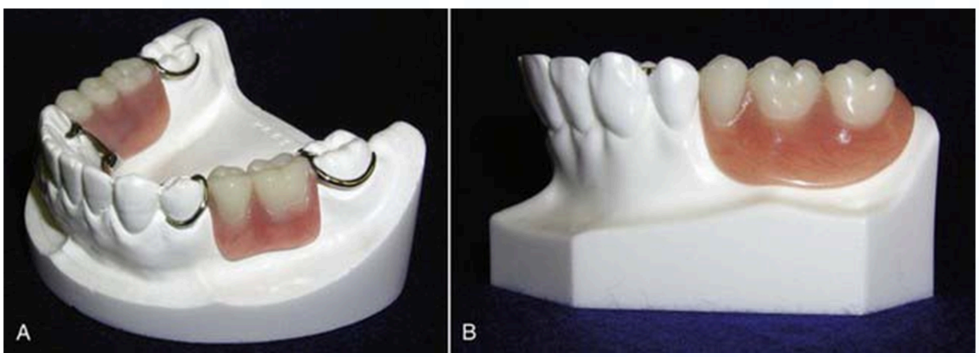



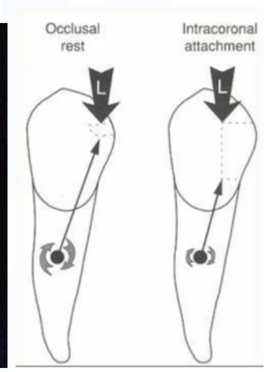




Advantages:

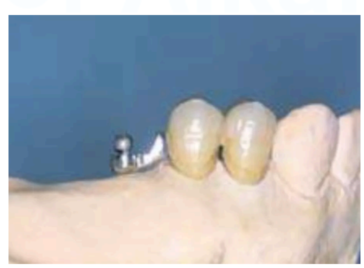
1. Improve esthetics by elimination of the visible retentive and support components.
2. Improve patient comfort.
3. Improve chewing efficiency.
4. Decreases the torque forces on abutment teeth.





Disadvantages:

1. It requires complicated clinical and laboratory work.
2. Large pulps may be jeopardised by the attachment component.
3. Difficult to repair and replace.
4. Less effective in short teeth.
5. More costly.
6. Requires high manual dexterity of the patient.



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6.2 Attachments for RPDs

Introduction

Attachments were developed after the turn of the twentieth century. Boitel reviewed this development beginning in 1915 when there were a few T-shaped and bar attachments. Terrell gives credit to Herman Chayes and B.B. McCollum for a major development in attachments and refers to them as "practice builders" and not as replacements for conventional removable partial dentures (Figure 6.2.1).

Figure 6.2.1. Intracoronal attachments incorporated into the castings with the RPD framework seated in position; intracoronal attachments eliminate the need for clasp assemblies.

Terms that describe attachments include precision and semiprecision, resilient and rigid, clips and snaps, key and keyway, matrix and patrix, and male and female. Although the terms male and female are popular, the terminology of matrix and patrix is preferred.

When considering fabrication of a removable partial denture with attachments, accuracy and precision are as important as when constructing any removable partial denture. For a metal removable partial denture that includes major connectors, minor connectors, rests, and direct retainers incorporated into a framework, the prosthesis must be well made to ensure a precise fit intraorally. Various types of attachments for use in removable partial dentures are available. The four categories of attachments include intracoronal attachments, extracoronal attachments, overdenture attachments, and bar-type attachments; each offers unique indications and advantages over the other types. However, there are also specific challenges and disadvantages in the use of each type of attachment.

Intracoronal attachments

Many manufacturers offer both cast metal attachments and components made of plastic patterns for custom fabrication. Intracoronal attachments are made with a key (patrix) and keyway (matrix) mechanism, typically manufactured such that the keyway or matrix fits within the contours of a crown and the key or patrix is a part of the removable partial denture framework (Figures 6.2.2 and 6.2.3). The patrix engages the vertical walls built within the contours of the crown and resists dislodgement by a torsional resistance of the metal. Intracoronal attachments may include locking mechanisms and frictional or spring retention. Another variation of an intracoronal attachment consists of a spring and plunger mechanism within the attachment in the removable partial denture and a depression into the normal contours of the interproximal surface on the abutment tooth (Figures 6.2.4 through 6.2.12).

Figure 6.2.2. Patrix sliding into matrix of an abutment crown.

Figure 6.2.3. Matrix-patrix mechanisms of intracoronal attachments are shown in abutment crowns.

Indications

Intracoronal attachments can help obtain esthetic results when used in lieu of traditional

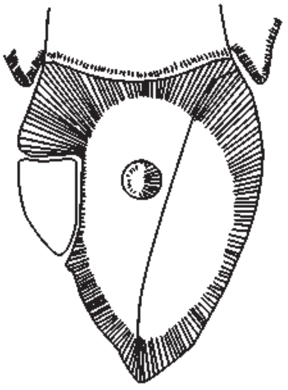

Figure 6.2.4. The depression in the proximal surface of the abutment tooth is shown in this diagram. The depression is seen as a uniform, circular depression.

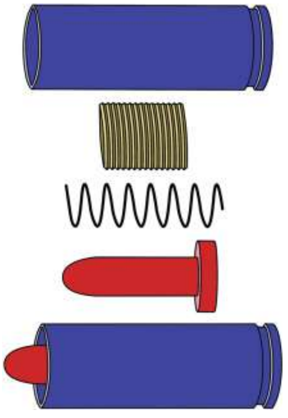

Figure 6.2.5. Plunger mechanism (patrix) of the attachment (top three diagrams) that will be incorporated into the removable partial denture framework. The assembled mechanism is shown at the bottom.

Figure 6.2.6. Abutment wax pattern is shown with the circular depression waxed into the proximal (distal) surface of the crown.

Figure 6.2.7. The casting shows a proximal depression that was waxed into the surface.

Figure 6.2.8. The metal ceramic crown has been completed and the view shows the plunger mechanism placed in position as it would be within the RPD framework.

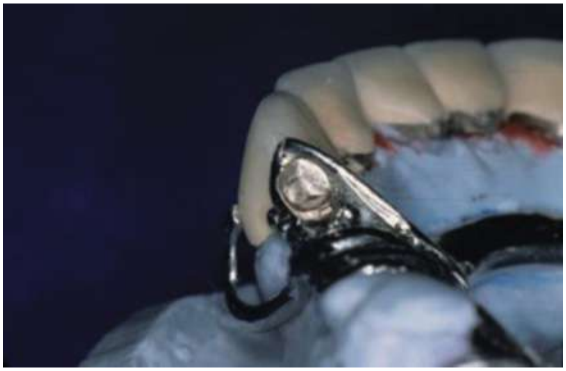

Figure 6.2.9. The plunger mechanism is distal to tooth no. 6, incorporated in the RPD framework.

Figure 6.2.10. Occlusal view of the RPD framework; the plunger mechanism is distal to tooth no. 6 and occupies a significant amount of space mesiodistally. Framework with plunger mechanism attached.

Figure 6.2.11. Tissue surface (intaglio) view of the completed RPD.

Figure 6.2.12. Completed RPD with attachment incorporated.

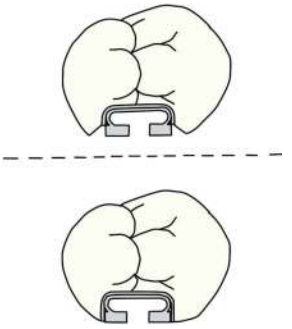

Figure 6.2.13. Diagram shows relative space needed to accommodate the matrix component of an intracoronal attachment within the confines of the extracoronal contours of a crown.

clasp assembly retention. An intracoronal attachment is indicated primarily for tooth-supported situations to provide esthetics and cross-arch stabilization.

Advantages

In addition to improving esthetic outcomes, intracoronal attachments provide improved leverage management. The attachment acts as a deep internal rest seat that transfers vertical forces closer to the axis of rotation on the abutment tooth. Intracoronal attachments also have a rigid connection that does not require indirect retention.

Disadvantages

The intracoronal attachment requires extensive preparation of an abutment tooth in order to obtain space for the matrix mechanism; aggressive crown preparation into the appropriate proximal surface allows creating a casting without creating an overcontoured crown, which could compromise periodontal health. This is particularly true on natural teeth that are narrow faciolingually, and also for teeth that have short clinical crowns. Based on this disadvantage, the large pulp chambers present in teeth of younger individuals present as a contraindication for use of intracoronal attachments (Figures 6.2.13 and 6.2.14).

Figure 6.2.14. Palatal view of an intracoronal attachment in an abutment crown that requires space for one of the attachment components.

A major problem with intracoronal attachments other than being difficult to repair is that the key or patrix wears over time, resulting in loss of retention. Patients also need to demonstrate good manual dexterity, since this type of prosthesis can be difficult to place and remove from the mouth.

Contraindications

Use of attachments is contraindicated in most mandibular distal-extension removable partial dentures (Kennedy Classifications I or II), especially when a type of stress director is not incorporated as part of the attachment. Functional movement of the prosthesis can generate stresses

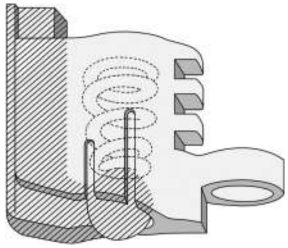

Figure 6.2.15. Spring-loaded extracoronal attachment with a vertical bar with ball incorporated into the distal surface of a crown as the patrix component, shaded gray. The matrix component is shown with an internal spring mechanism in housing that will be incorporated into the removable partial denture framework.

on the abutment tooth, and if not managed or designed well, can be excessive and result in loss of supporting bone.

Extracoronal attachments

Indications

Extracoronal attachments are equally as esthetic as intracoronal attachments, but unlike intracoronal attachments, they have the ability to provide more resilience as a stress director if this action is desired for a particular clinical situation (Figure 6.2.15).

Advantages

Use of extracoronal attachments can be perceived as easier for use as compared to intracoronal attachments. The extracoronal attachments are indicated frequently for anterior prostheses in younger patients who have large dental pulp chambers (Figures 6.2.16 and 6.2.17).

Disadvantages

Extracoronal attachments, by design, appear bulky outside the physiologic contours of the crown since more space is required within the removable partial denture. Many types have springs and component parts that can break or wear, requiring additional adjustment, repair, or replacement (Figure 6.2.18).

Figure 6.2.16. Extracoronal ring (matrix) extends proximally.

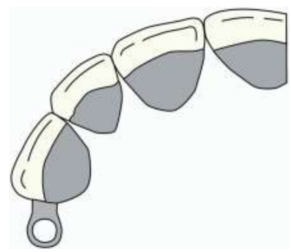

Figure 6.2.17. Diagram shows extracoronal attachment component that extends into the proximal area and requires space.

Overdenture attachments

Indications

As the title indicates, these types of attachments are used on overdenture abutments, either abutments that are natural teeth or dental implants. They are also referred to as stud-type attachments, include some of the lowest profile attachments in an occlusogingival dimension, and provide a stress-directing effect (Figure 6.2.19). The modes of attachment vary greatly among the various types and provide unique versatility. The individual types of attachments may vary as do the advantages and disadvantages of each type of overdenture attachment (Figure 6.2.20).

Figure 6.2.18. Attachment component must be accommodated within the RPD, which weakens this area of the prosthesis.

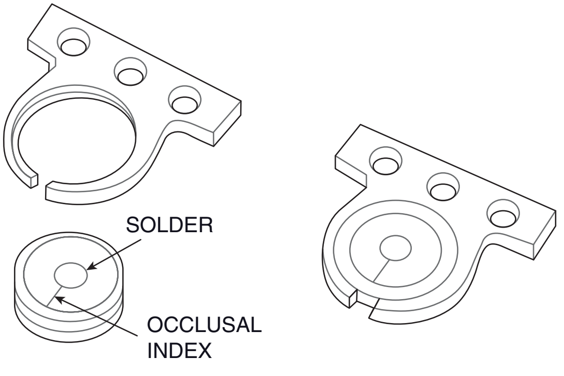

Figure 6.2.19. The Rotherman overdenture attachment is one of the lowest profile attachments, meaning it requires less occlusogingival space to accommodate in a prosthesis.

Advantages

In addition to versatility, advantages include tolerance when used with misaligned abutments, ease of maintenance, ease of adjustment, and repair. There is decreased leverage since the attachment mechanism closely approximates the axis of rotation when compared to the axis of rotation using the intracoronal attachments. The individual attachments selected should be evaluated and prescribed to maximize advantages on an individualized patient situation (Figures 6.2.21 through 6.2.23).

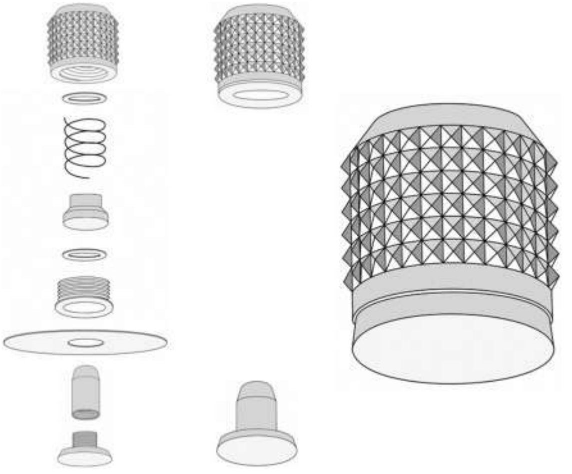

Figure 6.2.20. Design variability among attachment mechanisms provides versatility and choice in selection.

Figure 6.2.21. Ball component (patrix) for an O-ring attachment was cast to the overdenture abutment coping.

Figure 6.2.22. The O-ring and housing (matrix) have been placed on the ball attachment/abutment; a piece of rubber dam is inserted between the components as a blockout to prevent locking the mechanism during the clinical pickup procedure.

Figure 6.2.23. The view of the intaglio surface shows the O-ring incorporated into the removable partial denture.

Disadvantages

Clinical problems can occur, including adverse effects on prosthesis stability by the tilting effect or tipping potential, additional wear of mechanical components over time, design complexity for the attachment, available space from an occlusogingival perspective and space within the physiologic contours of the prosthesis, and additional expense incurred by the patient beyond the cost of the prosthesis.

Bar-type attachments

Most bar and clip attachments involve a type of mechanism in the design of the attachment (Figure 6.2.24). The differences in the mechanism are based on design to include clip retention, rotation around the bar, frictional retention of a superstructure, and combinations of clip and other types of attachments. Bar attachments can be cast, machine manufactured, milled, or refined using electrodischarge machining.

Advantages

A bar attachment provides advantages such as rigid splinting of natural teeth and cross-arch

Figure 6.2.24. Metal clip (matrix) snaps onto a bar.

Figure 6.2.25. Cross-arch stabilization is shown by use of a bar cast to abutment crowns on the two remaining mandibular canines.

stabilization. Indications include clinical scenarios in which there is considerable bone loss around the remaining abutment teeth (Figure 6.2.25). Other attachments may be used in conjunction with bars and/or in combination with osseointegrated implants for a combination fixed-removable prosthesis.

Disadvantages

The main problem with a bar attachment is restriction of use based on intraoral space limitations, interarch space, and interocclusal space, defined limits associated with ideal contours of a prosthesis. Additional laboratory procedures may be needed to fabricate a bar attachment

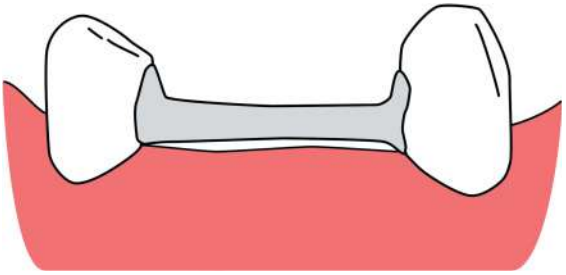

Figure 6.2.26. Minimal space gingivally exists between bar and tissue to reduce leverage forces.

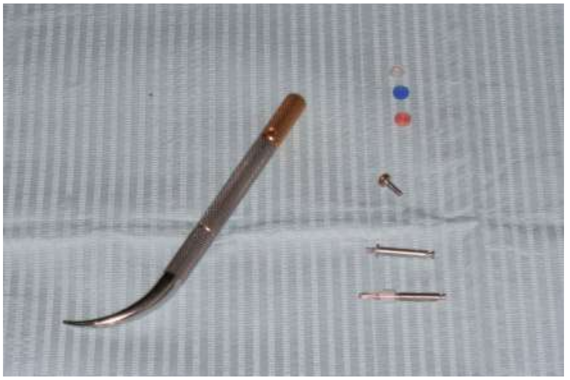

Figure 6.2.27. Custom components and instrumentation for use clinically and specifically for a type of an attachment.

such as soldering procedures, which can complicate the patient treatment (Figure 6.2.26). Plaque control is more difficult with a bar as compared to freestanding attachments, and the patient should have thorough home care instructions and appointments for oral hygiene maintenance procedures for long-term monitoring.

Fabrication of an overdenture attachment for a removable partial denture

As stated previously, there are different types of attachments, each with specific indications and advantages. Some require complicated and technique-sensitive dental laboratory procedures, while others are simple but involve meticulous attention to detail. It is not within the scope of this chapter to review the nuances of laboratory procedures among attachments; rather a technique that is used in a clinical setting is presented using one type of overdenture attachment.

Clinical Procedures

1. Examine the mouth. Complete clinical and radiographic examinations are necessary prior to prescribing the use of an attachment for a removable partial denture. A radiographic examination is valuable in viewing aspects of the abutment root and supporting structures. The clinical examination provides the evaluation of the remaining teeth and hard and soft tissues.

Figure 6.2.28. Dental materials, supplies, and instrumentation should be organized to ensure an efficient appointment for the practitioner and patient.

2. Obtain the appropriate instrumentation and components. Appropriate instruments must be identified while attachment-specific components should be selected and ordered (Figure 6.2.27).
3. Prepare for the clinical procedure. Many dental materials and supplies for these procedures are readily available in most dental offices (Figure 6.2.28).
4. Prepare the post space and cement the post manufactured with the patrix portion of the attachment. The patrix may also be waxed and cast onto a coping for a natural tooth, may be screwed into an implant, or may

Figure 6.2.29. The Locator™ matrix component can be cemented directly into the post space of a natural tooth abutment, eliminating the need for a casting.

even be manufactured as part of the implant (Figure 6.2.29).

5. Attach the nylon matrix to the patrix. Many attachment systems now include a metal housing as part of the system for ease of replacement. In the Locator™ system, the processing nylon matrix is black with essentially no retention. The terminology can be confusing, as the manufacturer of Locator™ calls a "male" or patrix what is referred to typically as a matrix. It may also be necessary to block out undercuts around the tooth or attachment using a wax or other product such as Oraseal™ (Ultradent, Salt Lake City, UT) (Figure 6.2.30).
6. Prepare the site for the matrix within the RPD. This is accomplished in many different ways, as some prefer a small vent hole and others prefer direct access through a larger hole in the prosthesis. In any case sufficient space must be available to accommodate the attachment.
7. Mix the autopolymerizing acrylic resin. This may be done by using either a salt and pepper technique with the RPD seated or by mixing the chemical-cured resin and placing the "free-flowing mix" into the prepared site within the RPD prior to seating the prosthesis (Figures 6.2.31 and 6.2.32).
8. Trim, finish, and polish the RPD. Excess resin is trimmed. The resin that extruded through the vent hole is removed and the

Figure 6.2.30. Housing and blockout spacer position on the natural tooth abutment.

Figure 6.2.31. Autopolymerizing resin is mixed and prepared for the clinical pickup procedure.

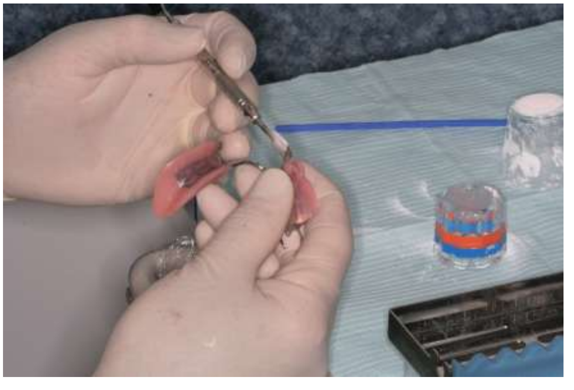

Figure 6.2.32. Resin is placed in the RPD.

Figure 6.2.33. RPD removed from the mouth.

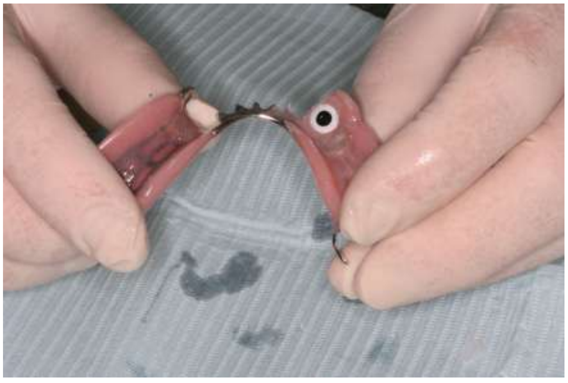

Figure 6.2.34. Intaglio view shows the nylon spacer incorporated into the RPD.

Figure 6.2.35. Excess resin is trimmed from the prosthesis.

Figure 6.2.36. Finish and polish the resin prior to insertion and delivery.

removable partial denture is polished (Figures 6.2.33 through 6.2.36).

9. Replace the black nylon processing matrix with a blue nylon matrix. The blue nylon matrix has the least amount of retention as described by the manufacturer. As the patient acclimates to the new attachment mechanism, subsequent nylon matrixes can be replaced with more retentive components (Figure 6.2.37).
10. Adjust the intaglio (tissue) surface. The tissue surface should be adjusted and relieved as needed using pressure indicator paste or a disclosing medium to identify areas that are potential sources of irritation for the patient.

Figure 6.2.37. The processing matrix is removed and replaced with the definite attachment into the removable partial denture.

Figure 6.2.38. Completed RPD.

11. Adjust the occlusion. The occlusion is checked with occlusal marking paper or ribbon to identify interferences and occlusal centric stops; the appropriate occlusion is established.
12. Review oral hygiene and home care of the RPD. Careful review of home care maintenance and basic insertion-removal techniques should be reviewed with the patient. A patient should be instructed to avoid "biting the RPD into place" in the mouth, as this could create problems with the attachment and unnecessary complications.
13. Postinsertion and maintenance: Patients should be seen within 24 hours after inserting the prosthesis to ensure patient comfort (Figure 6.2.38).

Summary

Four categories of attachments have been described: intracoronal, extracoronal, overdentures, and bar-type. With the predictable incorporation of osseointegrated implants into treatment planning a patient's needs, the realm of possibilities for removable partial dentures and attachments has increased. In clinical situations where a patient may have presented with too few teeth or poor locations of remaining abutment teeth, the addition of dental implants and various attachments can now be made into an excellent scenario for removable partial dentures. Although predictions based on insured populations may show less need for removable partial dentures, the reality is that with the development of new attachments and improved predictability of adjunctive dental implant placement, there should be an increased demand for alternative uses of adjunctive treatment.

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